

Instruction

You are an expert in database modeling.

Generate a Mermaid ER Diagram using ONLY valid Mermaid erDiagram syntax.

STRICT RULES

1. Use ONLY the entities, attributes, and relationships listed below.
2. Do NOT create additional entities.
3. Do NOT create additional attributes.
4. Do NOT create additional relationships.
5. Declare ALL entities before declaring ANY relationships.
6. Every entity referenced in a relationship must have an entity definition.
7. Output ONLY Mermaid code.
8. Do NOT include explanations.
9. Do NOT include markdown code fences.
10. Do NOT include quotes around relationship labels.

ATTRIBUTE FORMAT RULES

Use ONLY this format:

ENTITY_NAME { datatype ATTRIBUTE_NAME datatype ATTRIBUTE_NAME }

Example: SCHOOL { string SCHOOL_CODE string SCHOOL_NAME int PROF_NUM }

DO NOT USE:

SCHOOL { SCHOOL_CODE(string) SCHOOL_NAME(string) }

DO NOT USE: SCHOOL { SCHOOL_CODE PK, SCHOOL_NAME }

Every attribute must appear on its own line.

Valid datatypes:

- int
- string
- float
- date

RELATIONSHIP FORMAT RULES

Use ONLY this format: ENTITY1 ||--o{ ENTITY2 : relationship_name

Examples: SCHOOL ||--|{ DEPARTMENT : operates DEPARTMENT ||--|{ PROFESSOR :
employs PROFESSOR ||--o{ STUDENT : advises

DO NOT USE: ENTITY1 ||-- "relationship" ENTITY2 : one ENTITY1 -- ENTITY2 ENTITY1 :
relationship ENTITY2

DO NOT write:

- one
- many
- zero or many
- one or more inside relationship labels.

CARDINALITY RULES

Use ONLY these Mermaid cardinalities:

- || = exactly one
- o| = zero or one
- |{ = one or more
- o{ = zero or more

Never use:

- *{
- 1..N
- N:M

- UML notation

ENTITIES

SCHOOL Attributes: - SCHOOL_CODE - SCHOOL_NAME - PROF_NUM

DEPARTMENT Attributes: - DEPT_CODE - DEPT_NAME - SCHOOL_CODE - PROF_NUM

PROFESSOR Attributes: - PROF_NUM - DEPT_CODE - PROF_SPECIALTY - PROF_RANK -
PROF_LNAME - PROF_FNAME - PROF_INITIAL - PROF_EMAIL

STUDENT Attributes: - STU_NUM - DEPT_CODE - STU_LNAME - STU_FNAME -
STU_INITIAL - STU_EMAIL - PROF_NUM

COURSE Attributes: - CRS_CODE - DEPT_CODE - CRS_TITLE - CRS_DESCRIPTION -
CRS_CREDIT

CLASS Attributes: - CLASS_CODE - CLASS_SECTION - CRS_CODE - PROF_NUM -
ROOM_CODE

ENROLL Attributes: - CLASS_CODE - STU_NUM - ENROLL_DATE - ENROLL_GRADE

ROOM Attributes: - ROOM_CODE - ROOM_TYPE - BLDG_CODE

BUILDING Attributes: - BLDG_CODE - BLDG_NAME - BLDG_LOCATION

LIBRARY Attributes: - LIBRARY_ID - LIBRARY_NAME - LOCATION - HOURS

BOOK Attributes: - BOOK_ID - LIBRARY_ID - ISBN - TITLE - AUTHOR -
AVAILABILITY_STATUS

DORMITORY Attributes: - DORM_ID - DORM_NAME - CAPACITY - DORM_TYPE

STUDENT_HOUSING Attributes: - HOUSING_ID - STU_NUM - DORM_ID - ROOM_NUMBER
- MOVE_IN_DATE - MOVE_OUT_DATE

CLUB Attributes: - CLUB_ID - CLUB_NAME - MEETING_LOCATION - FACULTY_ADVISOR

CLUB_MEMBERSHIP Attributes: - MEMBERSHIP_ID - STU_NUM - CLUB_ID - JOIN_DATE -
ROLE

RELATIONSHIPS

A school operates many departments.

A department employs many professors.

A professor may serve as dean of a school.

A professor may chair one department.

Each department has one chair.

A department offers many courses.

A course generates many class sections.

A professor teaches many class sections.

A room is used for many class sections.

A building contains many rooms.

A department has many students.

A professor advises many students.

A student may have many enrollment records.

A class section may appear in many enrollment records.

A library contains many books.

A dormitory houses many housing assignments.

A student may have one housing assignment.

A professor may advise many clubs.

A student may join many club memberships.

A club may have many memberships.

IMPORTANT MODELING CONSTRAINTS

1. The academic portion of the ERD must closely resemble the classic Tiny College ERD.

2. ENROLL must connect ONLY STUDENT and CLASS1.

3. ENROLL must NOT connect to STUDENT_HOUSING.

4. LIBRARY connects ONLY to BOOK.

5. SCHOOL must NOT connect directly to LIBRARY.

6. BOOK must belong only to LIBRARY.
7. CLUB membership must be modeled through CLUB_MEMBERSHIP.
8. Do NOT create a direct STUDENT-to-CLUB relationship.
9. STUDENT_HOUSING must be used only between STUDENT and DORMITORY.
10. Preserve all original academic relationships from the Tiny College ERD.
11. Do not infer additional relationships beyond those explicitly listed.
12. Every relationship listed above must appear exactly once in the output.
13. Do not reverse relationship directions.
14. The final ERD must contain exactly 15 entities.

OUTPUT FORMAT

Start exactly with: erDiagram

Then declare ALL entities.

After all entities are declared, declare ALL relationships.

Output valid Mermaid ER Diagram syntax only.